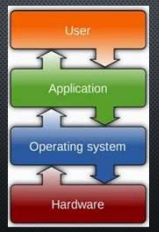
OPERATING SYSTEMS

OPERATING SYSTEMS

- A PROGRAM THAT MANAGES THE COMPUTER'S RESOURCES, PROVIDES SERVICES FOR PROGRAMMERS, AND SCHEDULES THE EXECUTION OF OTHER PROGRAMS.
- A PROGRAM THAT CONTROLS THE EXECUTION OF APPLICATION PROGRAMS AND **ACTS AS AN INTERFACE BETWEEN APPLICATIONS AND THE COMPUTER HARDWARE.**

OPERATING SYSTEMS



STRUCTURE OF OS

OPERATING SYSTEMS

TWO PRIMARY OBJECTIVE OF AN OS ARE:

1. MAKE A COMPUTER SYSTEM EASIER TO USE
2. MANAGE THE RESOURCES OF A COMPUTER SYSTEM

OPERATING SYSTEM SOFTWARE

OS SOFTWARE HAS FOUR MAJOR TYPES IN THE COMPUTER SYSTEM:

1. MEMORY MANAGEMENT
2. PROCESSOR MANAGEMENT
3. DEVICE MANAGEMENT
4. FILE MANAGEMENT

OPERATING SYSTEM SOFTWARE

OS SOFTWARE HAS FOUR MAJOR TYPES IN THE COMPUTER SYSTEM:

1. **MEMORY MANAGEMENT:**

- PRIMARY RESPONSIBILITY IS TO PROTECT THE SPACE IN THE MAIN MEMORY OCCUPIED BY THE OS ITSELF.
- IT CANNOT ALLOW ANY PART OF IT TO BE ACCIDENTALLY OR INTENTIONALLY ALTERED
- IT IS THE PROCESS OF CONTROLLING AND COORDINATING COMPUTER MEMORY, ASSIGNING PORTIONS CALLED BLOCKS TO VARIOUS RUNNING PROGRAMS TO OPTIMIZE OVERALL SYSTEM PERFORMANCE.

OPERATING SYSTEM SOFTWARE

2. PROCESSOR MANAGEMENT:

- IT DECIDES HOW TO ALLOCATE THE CPU TO KEEP TRACK OF THE STATUS OF EACH PROCESSES.
- IT MONITORS WHETHER THE CPU IS EXECUTING A PROCESS OF READ AND WRITE COMMAND
- IT HAS TWO LEVELS OF RESPONSIBILITY: JOB SCHEDULER AND PROCESS SCHEDULER

OPERATING SYSTEM SOFTWARE

3. DEVICE MANAGEMENT:

- MONITORS EVERY DEVICE, CHANNELS, AND CONTROL UNITS.
- IT CHOOSES THE MOST EFFICIENT WAY TO ALLOCATE ALL OF THE SYSTEM DEVICES (EXAMPLE: PRINTER, DISK DRIVES, ETC.)
- IT IS THE PROCESS OF MANAGING THE IMPLEMENTATION, OPERATION AND MAINTENANCE OF PHYSICAL AND/OR VIRTUAL DEVICE.

OPERATING SYSTEM SOFTWARE

4. FILE MANAGEMENT:

- KEEPS TRACK OF EVERY FILE IN THE SYSTEM WHICH INCLUDE DATA FILE, PROGRAM FILE, COMPILER, AND APPLICATION
- ENFORCES RESTRICTIONS ON WHO HAS ACCESS TO WHICH FILES
- CONTROLS WHAT ARE ALLOWED TO DO WITH THE FILES ONCE THEY ARE ACCESSED